In [1]:
```
# Assignment-5
# PRN: 260240128008                          PRN: 260240128036
# Name: Ashlesha Tayade                      Name: Ruchika Gaikwad
```

In [2]:
```
'''Topic: Matplotlib

1.      Draw line chart to show urban and rural population of India over the years using population.csv [rural  p
2.      Using a Bar chart display India's total and rural population from year 2020 to 2025
3.      Using gapminder2007.csv show life expectancy of top 10 highly populated countries from Asia
4.      Using a pie chart show distribution of top six selling cars [Create data as required]
5.      Using Histogram show distribution of developers by their experience [use survey_data_sample.csv]

Note: use appropriate labels, title and legend for each chart'''
```

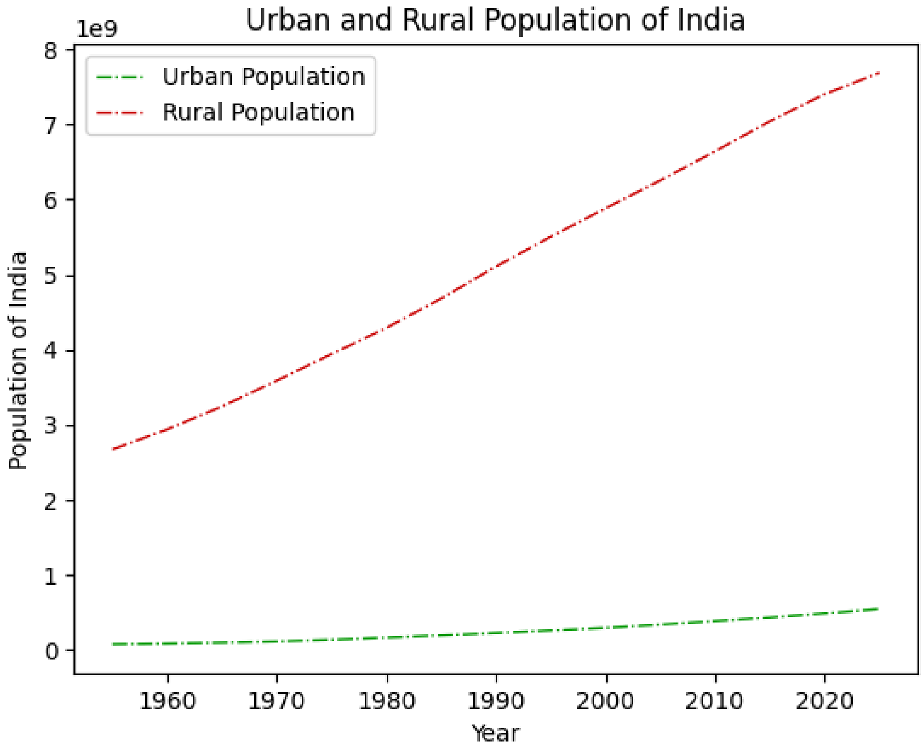
In [3]:
```
import pandas as pd
from matplotlib import pyplot as plt
import numpy as np
```

In [4]:
```
# 1.Draw line chart to show urban and rural population of India over the years using population.csv
# [rural population is the difference between total population and urban population]
df=pd.read_csv("Population.csv")
df
```

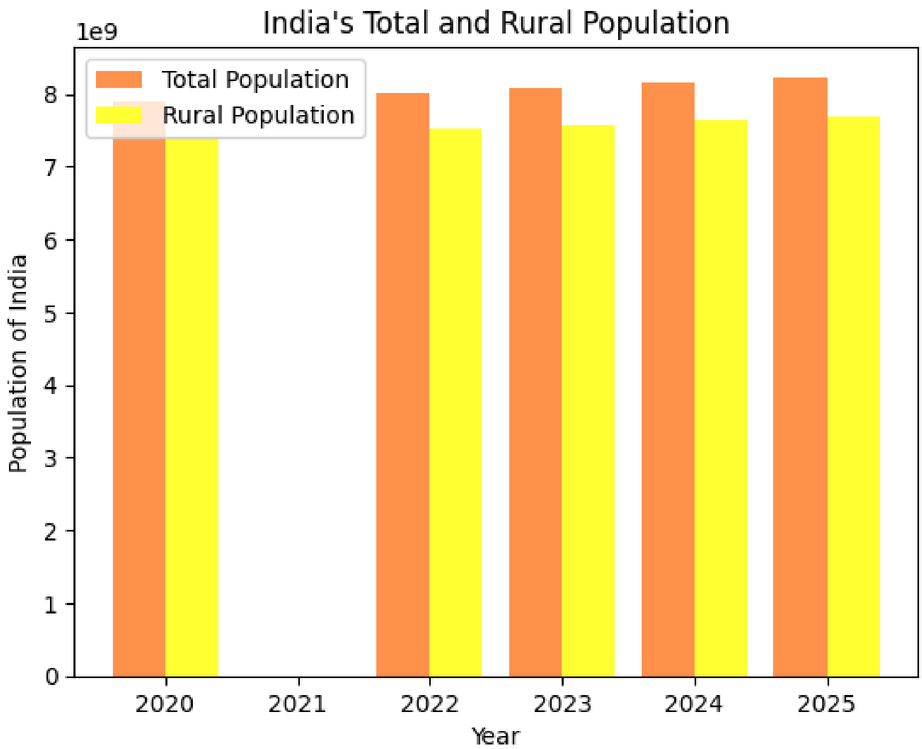
Out[4]:

	Year	Population	Yearly % Change	Yearly Change	Median Age	Fertility Rate	Density	Urban Pop %	Urban Population	Country's Share	World Population
0	1955	3.877009e+08	2.29%	8284413.0	19.7	5.91	130	18.60%	71958495.0	14.15%	2.740214e+09
1	1960	4.359903e+08	2.38%	9657890.0	19.2	5.92	147	18.50%	80565723.0	14.46%	3.015471e+09
2	1965	4.901401e+08	2.37%	10829962.0	18.5	5.94	165	19.10%	93493844.0	14.70%	3.334534e+09
3	1970	5.458643e+08	2.18%	11144824.0	18.1	5.62	184	20.00%	109388950.0	14.77%	3.694684e+09
4	1975	6.113095e+08	2.29%	13089053.0	18.4	5.20	206	21.70%	132533810.0	15.02%	4.070735e+09
5	1980	6.873540e+08	2.37%	15208898.0	18.9	4.78	231	23.40%	160941941.0	15.45%	4.447606e+09
6	1985	7.726478e+08	2.37%	17058754.0	19.3	4.43	260	24.60%	190321782.0	15.87%	4.868943e+09
7	1990	8.649722e+08	2.28%	18464886.0	19.7	4.04	291	25.70%	222296728.0	16.24%	5.327803e+09
8	1995	9.603010e+08	2.11%	19065765.0	20.3	3.65	323	26.60%	255558824.0	16.68%	5.758879e+09
9	2000	1.057923e+09	1.96%	19524338.0	21.2	3.35	356	27.50%	291350282.0	17.14%	6.171703e+09
10	2005	1.154676e+09	1.77%	19350718.0	22.2	2.96	388	29.00%	334479406.0	17.53%	6.586970e+09
11	2010	1.243482e+09	1.49%	17761048.0	23.6	2.60	418	30.60%	380744554.0	17.71%	7.021732e+09
12	2015	1.328024e+09	1.32%	16908587.0	25.3	2.29	447	32.30%	429069459.0	17.78%	7.470492e+09
13	2020	1.402618e+09	1.10%	14918639.0	27.0	2.05	472	34.40%	483098640.0	17.78%	7.887001e+09
14	2022	1.425423e+09	0.81%	11402759.0	27.7	1.99	479	35.50%	506304869.0	17.77%	8.021407e+09
15	2023	1.438070e+09	0.89%	12646384.0	28.1	1.98	484	36.00%	518239122.0	17.77%	8.091735e+09
16	2024	1.450936e+09	0.89%	12866195.0	28.4	1.96	488	36.60%	530387142.0	17.78%	8.161973e+09
17	2025	1.463866e+09	0.89%	12929734.0	28.8	1.94	492	37.10%	542742539.0	17.78%	8.231613e+09

In [5]:
```
plt.plot(df["Year"],df["Urban Population"], color="#009900" ,lw=1,ls="-.", label="Urban Population")
plt.plot(df["Year"],df["World Population"]-df["Urban Population"], color="#cc0000" ,lw=1,ls="-.", label="Rural Po
plt.xlabel("Year")
plt.ylabel("Population of India")
plt.legend()
plt.title("Urban and Rural Population of India")
plt.show()
```



In [6]:
```
# 2. Using a Bar chart display India's total and rural population from year 2020 to 2025
filt=df["Year"] .isin([2020,2021,2022,2023,2024,2025])
year=df.loc[filt,:]
width=0.4
plt.bar(year["Year"]-width/2,year["World Population"], width=width, color="#ff944d" , label="Total Population")
plt.bar(year["Year"]+width/2,year["World Population"]-year["Urban Population"], width=width,color="#ffff33", labe
plt.xlabel("Year")
plt.ylabel("Population of India")
plt.legend()
plt.title("India's Total and Rural Population")
plt.show()
```



In [7]:
```
# 3.Using gapminder2007.csv show life expectancy of top 10 highly populated countries from Asia
df2 =pd.read_csv("gapminder2007.csv")
df2
```

Out[7]:

	country	pop	continent	lifeExp	gdpPercap
0	Afghanistan	31889923	Asia	43.828	974.580338
1	Albania	3600523	Europe	76.423	5937.029526
2	Algeria	33333216	Africa	72.301	6223.367465
3	Angola	12420476	Africa	42.731	4797.231267
4	Argentina	40301927	Americas	75.320	12779.379640
...	...	...	...	...	...
137	Vietnam	85262356	Asia	74.249	2441.576404
138	West Bank and Gaza	4018332	Asia	73.422	3025.349798
139	Yemen, Rep.	22211743	Asia	62.698	2280.769906
140	Zambia	11746035	Africa	42.384	1271.211593
141	Zimbabwe	12311143	Africa	43.487	469.709298

142 rows × 5 columns

In [8]:

```
filt =df2["continent"]=="Asia"
country_data = df2.loc[filt,:]
data=country_data.nlargest(10,"pop")
data
```

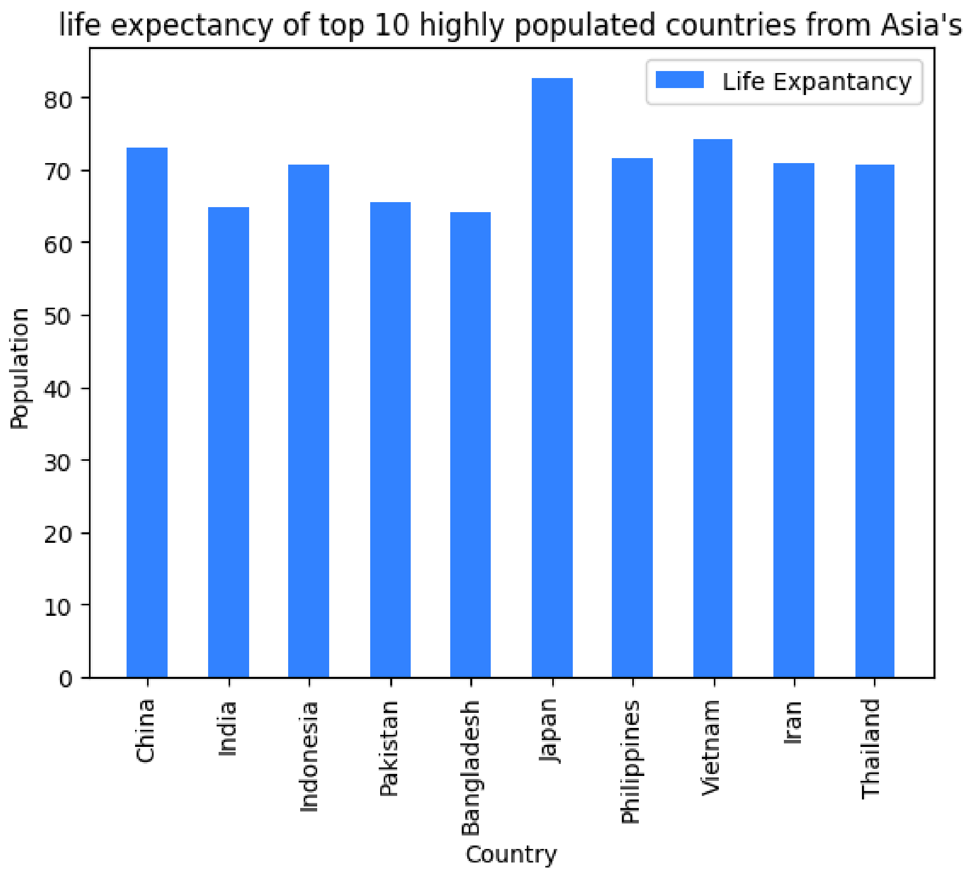
Out[8]:

	country	pop	continent	lifeExp	gdpPercap
24	China	1318683096	Asia	72.961	4959.114854
58	India	1110396331	Asia	64.698	2452.210407
59	Indonesia	223547000	Asia	70.650	3540.651564
97	Pakistan	169270617	Asia	65.483	2605.947580
8	Bangladesh	150448339	Asia	64.062	1391.253792
66	Japan	127467972	Asia	82.603	31656.068060
101	Philippines	91077287	Asia	71.688	3190.481016
137	Vietnam	85262356	Asia	74.249	2441.576404
60	Iran	69453570	Asia	70.964	11605.714490
127	Thailand	65068149	Asia	70.616	7458.396327

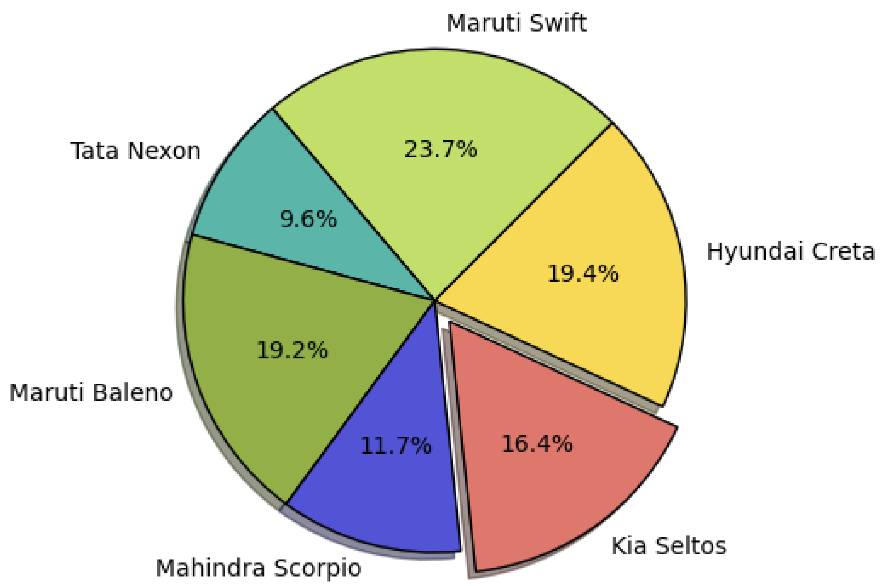
In [9]:

```
plt.bar(data["country"],data["lifeExp"], width=0.5, color="#3385ff" , label="Life Expantancy")
plt.xlabel("Country")
plt.ylabel("Population")
plt.xticks(rotation=90)
plt.legend()
plt.title("life expectancy of top 10 highly populated countries from Asia's")
```

Out[9]: Text(0.5, 1.0, "life expectancy of top 10 highly populated countries from Asia's")



In [10]:
```
# 4.Using a pie chart show distribution of top six selling cars [Create data as required]
Cars =["Maruti Swift", "Tata Nexon", "Maruti Baleno", "Mahindra Scorpio", "Kia Seltos", "Hyundai Creta"]
Sales= [11100000,4500000,9000000,5500000,7700000,9100000]
plt.pie(Sales, labels = Cars,
       colors =['#C2DE6B','#5BB5A9',"#92B047",'#5353D5','#DE786D','#F7D957'],
       startangle=45,
        explode=[0, 0, 0, 0, 0.1, 0],
       shadow=True,
       wedgeprops={'edgecolor':'Black'},
       autopct="%1.1f%%")
plt.show()
```



In [11]:
```
# 5.Using Histogram show distribution of developers by their experience [use survey_data_sample.csv]
developers_data =pd.read_csv("survey_results-sample.csv")
developers_data
```

Out[11]:

	ResponseId	Age	YearsCode	CompTotal	ConvertedCompYearly
0	1	18-24 years old	NaN	NaN	NaN
1	2	25-34 years old	18	2.850000e+05	285000.0
2	3	45-54 years old	27	2.500000e+05	250000.0
3	4	25-34 years old	12	1.560000e+05	156000.0
4	5	25-34 years old	6	1.320000e+06	23456.0
...	...	...	...	...	...
89179	89180	25-34 years old	20	2.000000e+05	NaN
89180	89181	18-24 years old	5	NaN	NaN
89181	89182	Prefer not to say	10	NaN	NaN
89182	89183	Under 18 years old	3	NaN	NaN
89183	89184	35-44 years old	17	3.300000e+09	NaN

89184 rows × 5 columns

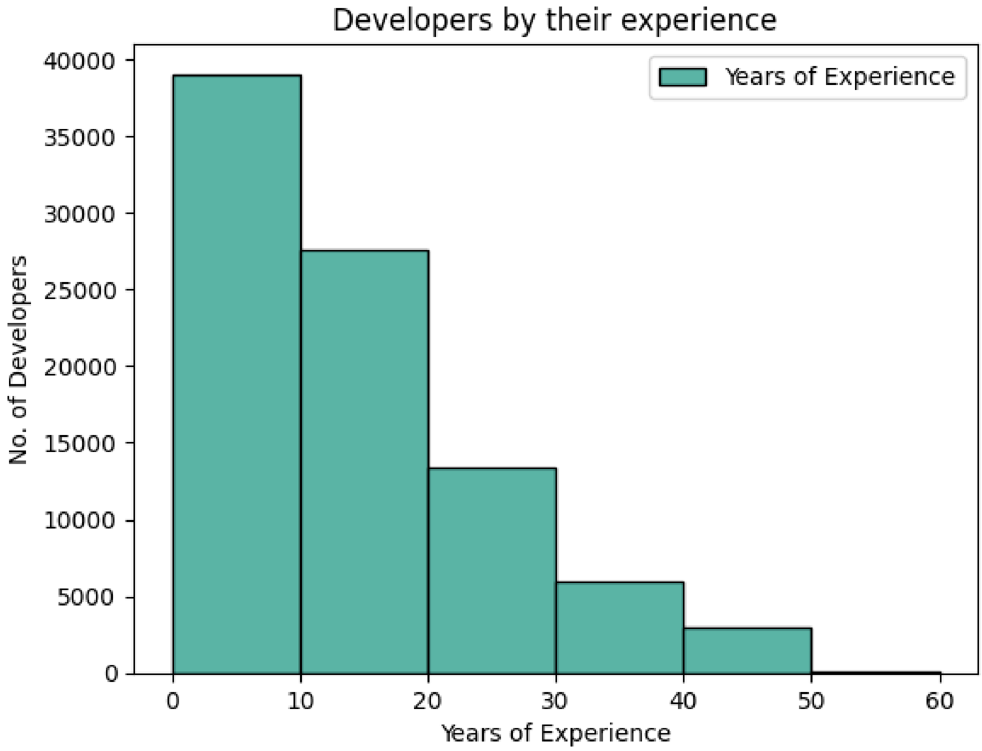
In [12]:
```
developers_data["YearsCode"]=developers_data["YearsCode"].fillna(0)
developers_data
```

Out[12]:

	ResponseId	Age	YearsCode	CompTotal	ConvertedCompYearly
0	1	18-24 years old	0	NaN	NaN
1	2	25-34 years old	18	2.850000e+05	285000.0
2	3	45-54 years old	27	2.500000e+05	250000.0
3	4	25-34 years old	12	1.560000e+05	156000.0
4	5	25-34 years old	6	1.320000e+06	23456.0
...	...	...	...	...	...
89179	89180	25-34 years old	20	2.000000e+05	NaN
89180	89181	18-24 years old	5	NaN	NaN
89181	89182	Prefer not to say	10	NaN	NaN
89182	89183	Under 18 years old	3	NaN	NaN
89183	89184	35-44 years old	17	3.300000e+09	NaN

89184 rows × 5 columns

In [64]:
```
bins=[0,10,20,30,40,50,60]
plt.hist(developers_data["YearsCode"], bins=bins, color="#5BB5A9",edgecolor="Black", label="Years of Experience")
plt.xlabel("Years of Experience")
plt.ylabel("No. of Developers")
plt.legend()
plt.title("Developers by their experience")
plt.show()
```



Developers by their experience
Years of Experience
40000
35000
30000
25000
20000
15000
10000
5000
0
No. of Developers
0
10
20
30
40
50
60
Years of Experience

In [ ]: